

FULLY ADAPTABLE
ADULTS

FULLY ADAPTABLE FIT
Fits most wrist sizes
(7"–9 ½")

Super Soft silicone
material provides
comfortable fit

Fits most wrist
sizes (7"–9 ½")

Share your details
via **NFC Tag**

Or Share your details
via **QR code** scan

Smart Emergency Response

The Smart Emergency Response :bracelet is an advanced healthcare device designed to monitor vital signs in real-time. It utilizes high-precision sensors to track heart rate, blood pressure, oxygen levels, and body temperature, providing continuous health monitoring. The system is equipped with intelligent alerts that offer personalized instructions or medication reminders for minor changes in vital signs. In critical situations, it can automatically send emergency messages to nearby hospitals, including GPS location data, ensuring rapid medical intervention. It also includes a manual SOS option, allowing users to call for help instantly if needed.

smart emergency response

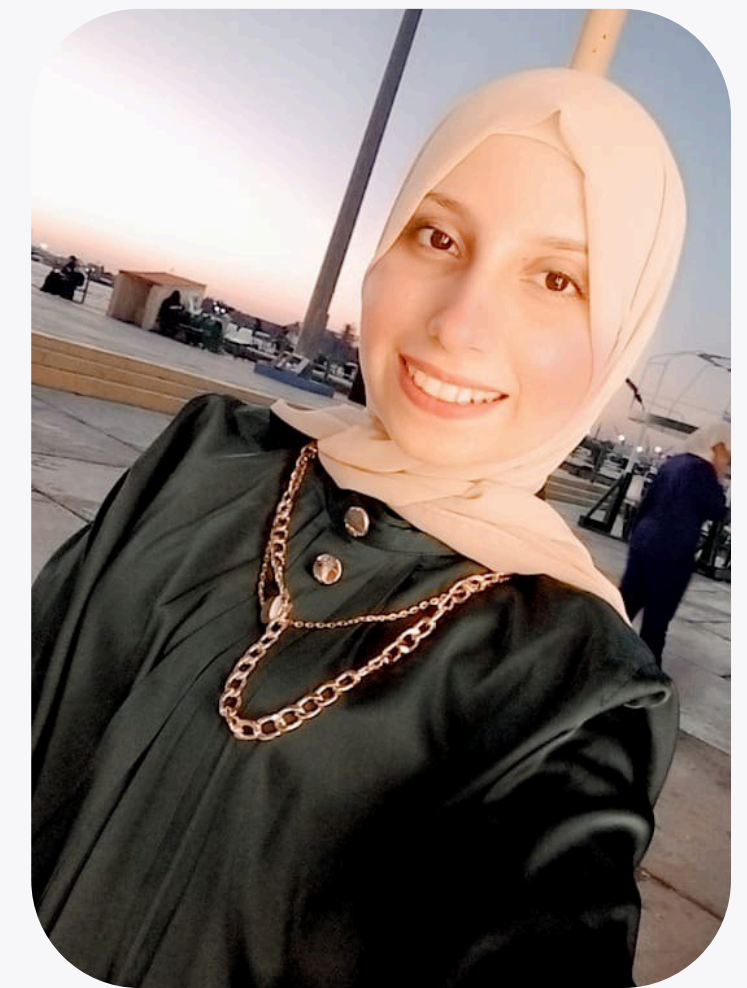
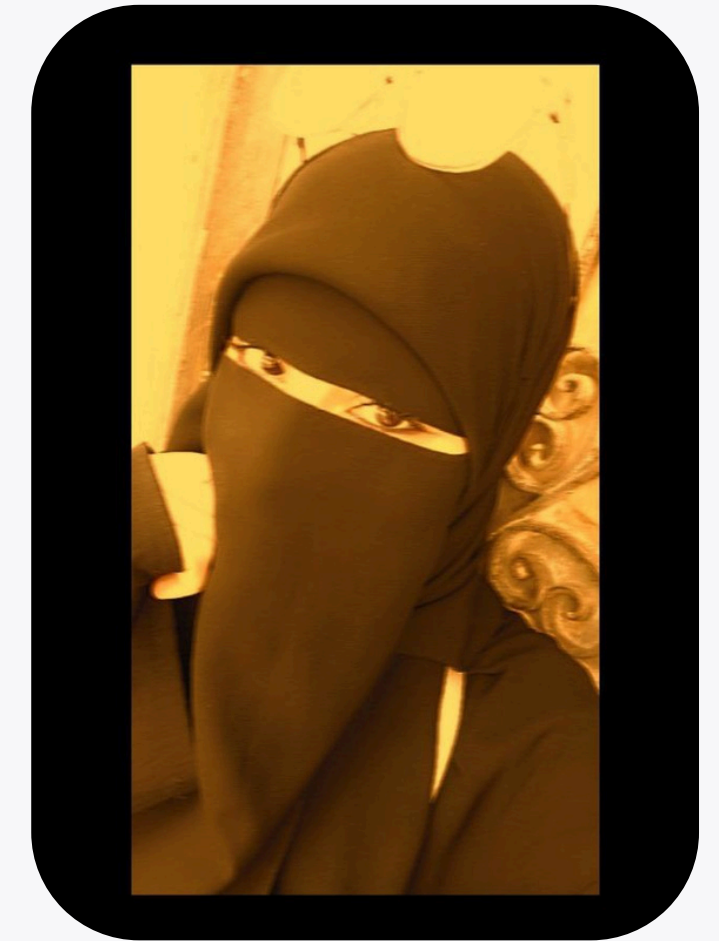
ADMINISTRATOR: DR.
NASMAA

ROUFIDA MOHAMED
AWADEEN

ALAA AHMED MOSAD

MENNA ABDO ELKEFAFY

NASSMA KHAIRY ELSAYED



System Title

Smart Emergency Medical Response System

Problem Statement & Solution :

Medical emergencies require immediate attention and delays in response can lead to severe health complications or death. Traditional emergency systems often suffer from slow communication, limited resource tracking, and delayed dispatching.

The Smart Emergency Medical Response System addresses these issues through :
Continuous monitoring of patient vital signs via wearable devices.

Automatic detection of critical conditions (e.g., heart attacks, strokes).

Instant alerts to the nearest ambulance and hospital.

Real-time patient location tracking using GPS.

Seamless communication between patients, emergency responders, and hospitals.



System Objective:

To reduce emergency response times and enhance the quality of urgent medical care by leveraging smart technologies, ultimately saving lives and improving patient outcomes.

Primary System Users:

Patients & Relatives: Request emergency assistance and receive medical alerts.

Ambulance Teams: Respond to alerts, provide on-site care, and transport patients.

Hospital Staff: Prepare for incoming cases, monitor alerts, and update facility status.

Health Administrators: Oversee system performance, analyze response data.

Functional Requirements

For Patients:

- Buy/Rent smart medical bracelet.
- Register to the system.
- Manually request emergency assistance.
- Acknowledge alerts received
- Subscribe to premium services and pay corresponding fees.
- Receive treatment reminders and emergency instructions.
- Provide feedback about system performance.
- Block access in case of misuse. (Extension)

For Ambulance Teams

- Receive patient location and instructions.
- Provide immediate on-site care.
- Transport patient to designated hospital.
- Update arrival and case status upon completion

For Hospital Staff

- Monitor incoming emergency alerts.
- Track real-time patient status and location.
- Update hospital capacity and lock hospital in case of unavailability.

(Extension)

- Notify ambulance team about hospital readiness.
- Subscribe to system and manage service access.

Non-Functional Requirements

- **Performance**

Emergency alerts must be delivered within 3 seconds.

System should support thousands of users concurrently with no degradation in performance.

- **Security & Privacy**

Role-based access control (RBAC) for all users.

End-to-end encryption for sensitive medical and personal data

Multi-factor authentication for access to hospital and admin systems.

- **Availability & Reliability**

System must be operational 24/7, with minimal downtime.

Redundant backup servers to ensure high availability in all regions.

- **Usability**

Intuitive user interface designed for all age groups.

Multi-language support with accessibility features (e.g., text-to-speech, large fonts).

- **Scalability**

Modular infrastructure to allow nationwide expansion and international deployment.

- **Accuracy**

Smart algorithms to minimize false alarms and ensure reliable emergency detection.

USER REQUIREMENTS

- Users must be able to request help through a wearable bracelet or mobile app.
 - Premium users can access additional features like medication reminders and priority dispatch.
 - Feedback submission should be available post-incident.
 - Ambulance teams need access to live patient data and GPS navigation.
-
- Hospital staff must be able to monitor, update, and lock hospital availability.
 - Health officials require system-wide analytics dashboards and reports.

System Requirements

- Integration with hospital databases to access bed availability and emergency resources.
- GPS tracking integration for accurate location dispatch.
- Payment gateway for premium service subscription.
- Admin panel for user management, alert logs, and performance metrics.
- Support for mobile platforms (Android/iOS) and web portals.

Project Sponsors

- **Primary:** Ministry of Health, National Emergency Services Authority.
- **Secondary:** Private medical technology investors and innovation funds.

BUSINESS REQUIREMENTS

- Launch a national network of wearable emergency devices.
- Develop scalable cloud-based application for real-time coordination.
- Implement strong security and compliance with healthcare laws (e.g., HIPAA).
- Provide training for ambulance and hospital personnel.
- Ensure integration with existing healthcare systems.

Special Constraints

- **Budgetary Constraints:** Project must offer cost-effective hardware/software options.
- **Connectivity Limitations:** System should support low-bandwidth and offline alert fallback.
- **Legal Compliance:** Must adhere to data privacy and medical emergency laws
- **Cultural Sensitivity:** System should accommodate diverse populations and accessibility needs.
- **Geographical Limitations:** Rural and remote areas must be fully supported.

Business Need:

Delays in medical response can lead to loss of life in critical conditions such as heart attacks or strokes.

Lack of coordination between patients, ambulances, and hospitals hinders effective emergency response.

Over-reliance on human interaction in emergencies increases the risk of delay or error, especially if the patient is unconscious.

Absence of real-time patient vital signs data makes it difficult to provide immediate and accurate medical care.

Need to enhance the efficiency and prioritization of emergency case management through smart technology.

Business Value:

Reduces mortality rates by enabling rapid response to medical emergencies. Improves efficiency of emergency services through automated communication between devices, ambulances, and hospitals. Saves time and cost by allowing hospitals to prepare in advance for incoming critical cases. Enhances patient and family trust by providing reliable and instant emergency support. Provides continuous and accurate patient data that helps in making timely medical decisions. Scalable and replicable system model that can be implemented in other cities or countries, offering high investment potential. Positions the developing organization as a leader in health-tech innovation, opening doors for partnerships and growth.

Conducting a feasibility study for the Economy Card is a crucial step to ensure the financial viability of this key component in the Smart Emergency Medical Response System. This analysis aims to assess the potential profitability and long-term economic sustainability of the project. It covers the expected revenues and associated costs, focusing on critical financial metrics such as Net Cash Flow, Cumulative Net Cash Flow Break Even point (BEP) and Net Present Value (NPV). These indicators provide a comprehensive view of the project's ability to generate positive returns, recover initial investments, and contribute to long-term financial stability so we did these steps.

	Year 0	Year 1	Year 2	Year 3	TOTAL
Benefits:					
Device Sales		2, 500, 000	2, 750, 000	3, 225, 000	8,475,000
Device rental		1, 080, 000	1, 880, 000	2, 306, 800	5,266,800
Hospital/Clinic Subscriptions		500, 000	900, 000	1, 260, 000	2,660,000
Premium Monitoring Services		720, 000	792, 000	871, 200	2,383,200
Total benefits		4, 800, 000	6, 322, 000	7, 663, 000	18,785,000
Costs:					
Manufacturing Hardware	1, 100, 000	1, 210, 000	1, 211, 000	1, 333, 300	4,854,300
Software Development	200, 000	220, 000	250, 000	290, 000	960,000
Marketing	40, 000	44, 000	49, 000	50, 000	183,000
Support & Maintenance	60, 000	66, 000	70, 000	74, 000	270,000
Bracelet R&D and Upgrades		60, 000	77, 000	85, 000	222,000
Total costs	1, 400, 000	1, 600, 000	1, 657, 000	1, 832, 300	6,489,300
Total benefits - Total costs	[1, 400, 000]	3, 200, 000	4, 665, 000	5, 830, 700	12,295,700
Comulative net cash flow	[1, 400, 000]	1, 800, 000	6, 465, 000	12, 295, 700	

Break-Even Point (BEP)

The break-even point (also called the payback method) is defined as the number of years it takes a firm to recover its original investment in the project from net cash flows.

It helps determine how long the project will take to reach a point where it starts making profit.

$$\text{BEP} = \text{Number of years of negative cash flow} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

$$= 0 + \frac{3,200,000 - 1,800,000}{1,800,000} = 0.78 \text{ years}$$

This means the project is expected to break even after approximately **8 months**, after which it starts generating net profit..

Present value(pv):

The current value of future money, discounted at a specific interest rate to reflect the time value of money.

Formula:

$$PV = \frac{\text{cash flow amount}}{(1 + \text{rate of return})^n}$$

	Year 0	Year 1	Year 2	Year 3	total
Total Benefits		4, 800, 000	6, 322, 000	7, 663, 000	18, 785, 000
Pv of total Benefits		4, 037, 005	4, 471, 886	4, 558, 828	13, 067, 719
Total costs	1, 400, 000	1, 600, 000	1, 657, 000	1, 832, 300	6, 489, 300
Pv of total costs	1, 400, 000	1, 345, 668	1, 172, 084	1, 090, 061	5, 007, 813

Net present value (NPV):

The NPV is simply the difference between the total present value of the benefits and the total present value of the costs

$$NPV = \sum PV \text{ of total Benefits} - \sum pv \text{ of total costs}$$

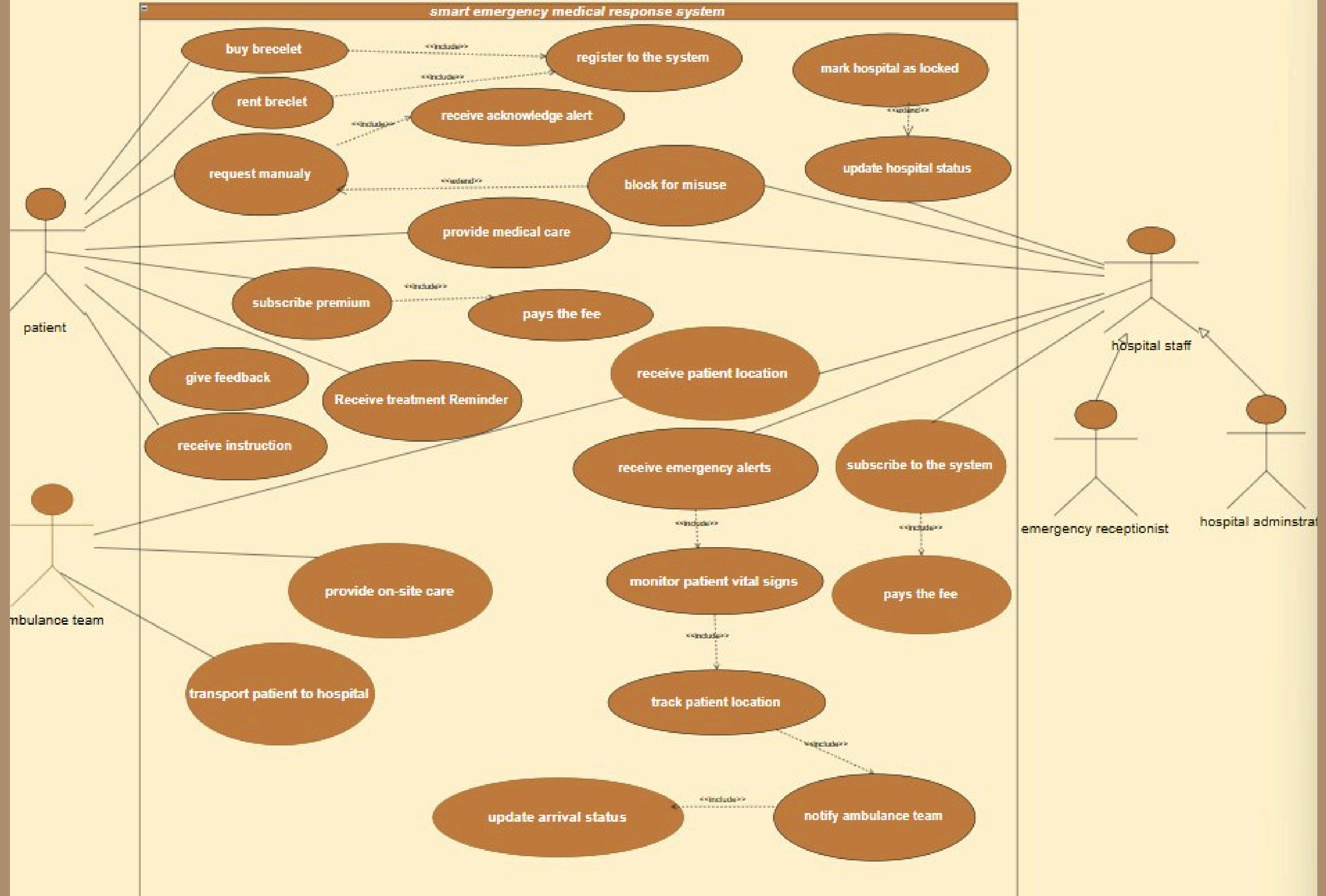
$$=\$ 13,067,719 - \$ 5,007,813 = \$ \mathbf{8,059,906}$$

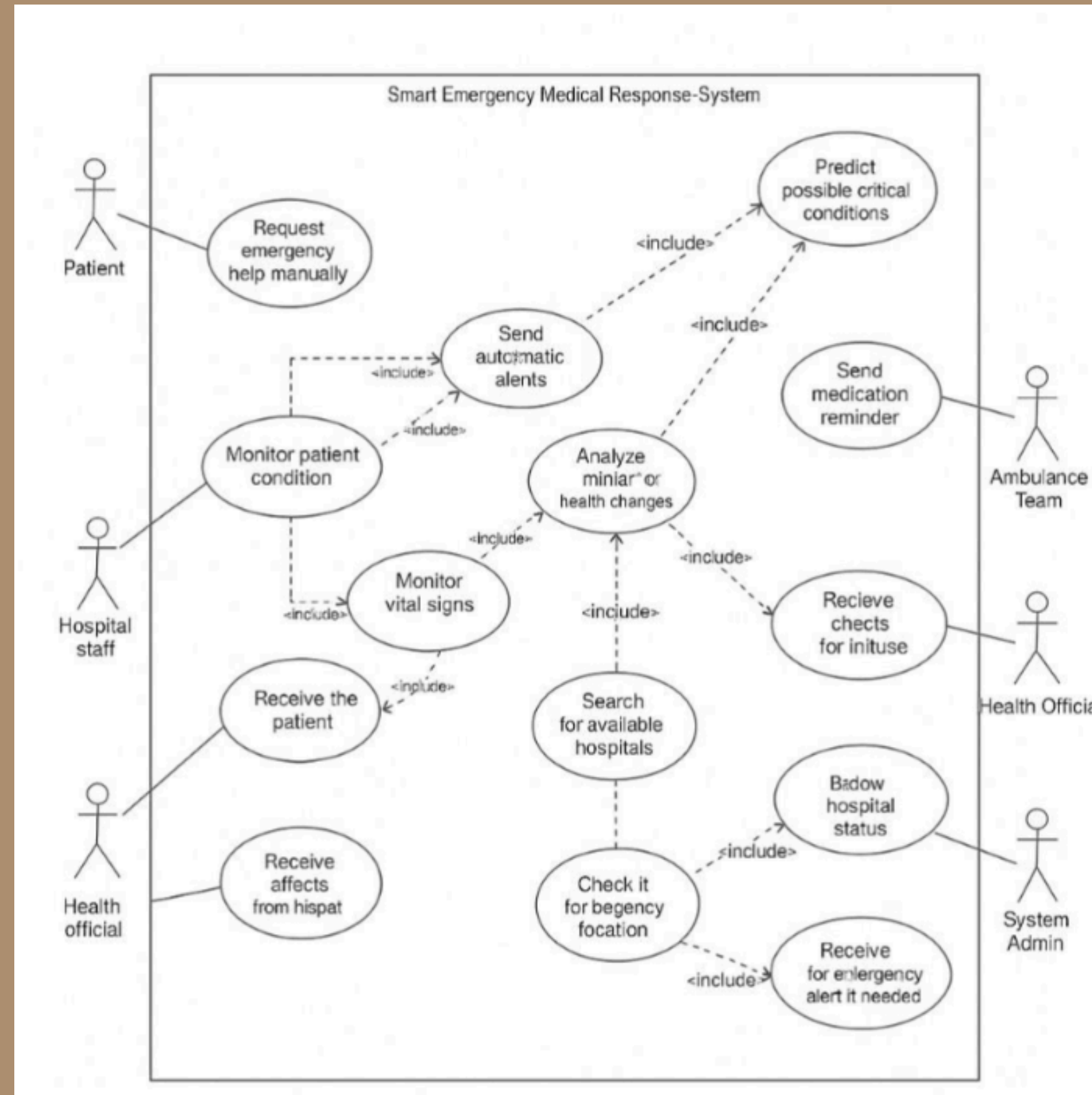
As we see NPV is **greater than Zero** so this project is considered **economically acceptable** .

Use Case daigram:

The use case defines the core interactions between the user and the Smart Emergency Medical Response System. It outlines how the user, wearing the Economy Card bracelet, interacts with the system during emergencies. This includes sending real-time alerts, accessing vital medical information, and ensuring rapid response through seamless data sharing. The use case captures critical processes like emergency detection, data transmission, and location tracking to enhance patient safety and improve response times.



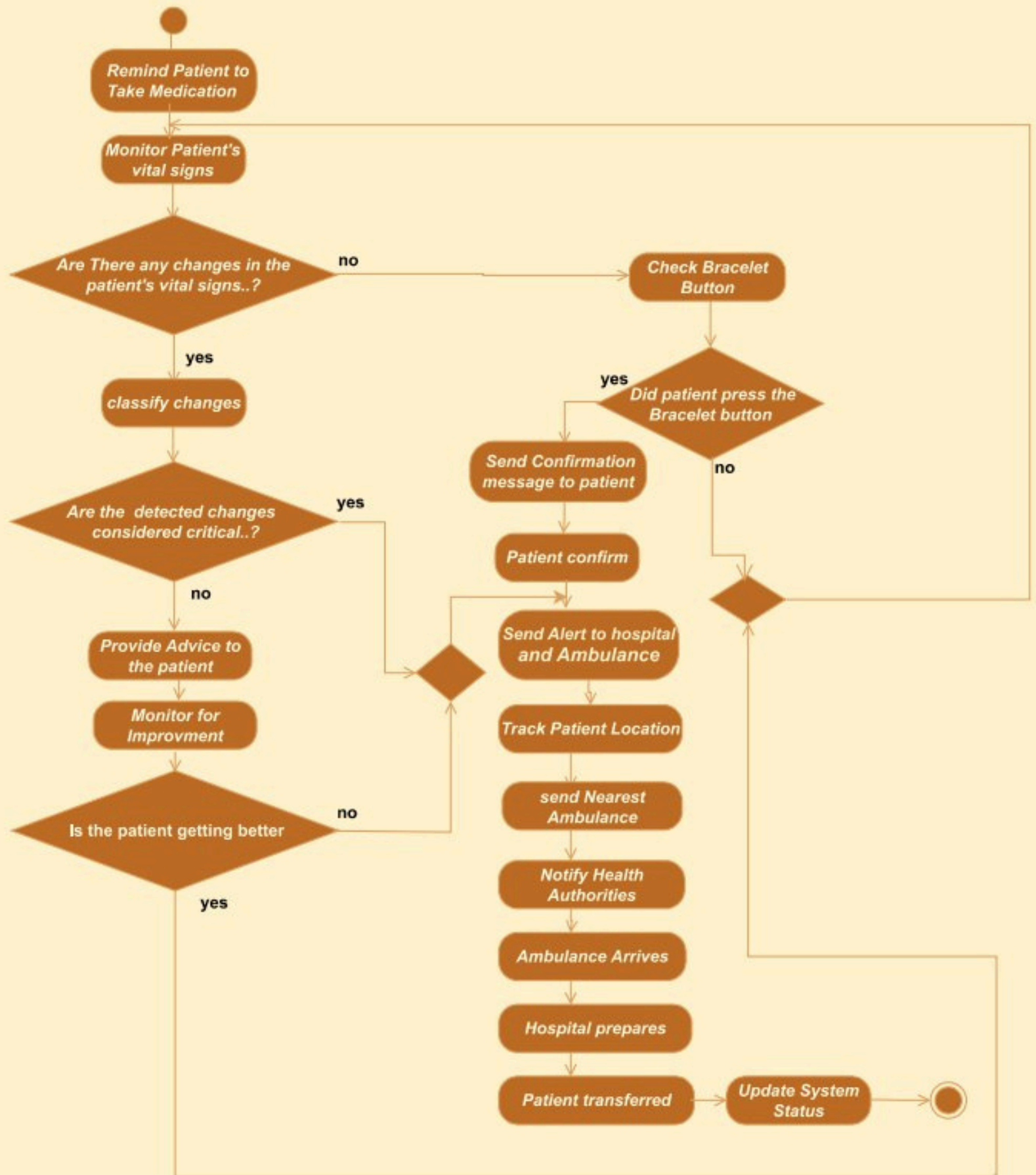




Activity diagram

"The activity diagram is a crucial tool for modeling the flow of activities within the Smart Emergency Medical Response System. It visually represents the sequence of steps that occur from the moment an emergency is detected until the system sends alerts and coordinates the response. The diagram outlines actions such as patient data transmission, emergency notification, location tracking, and system response. By illustrating these processes, the activity diagram helps to understand the interaction between different system components and ensures smooth and efficient operations in critical situations."





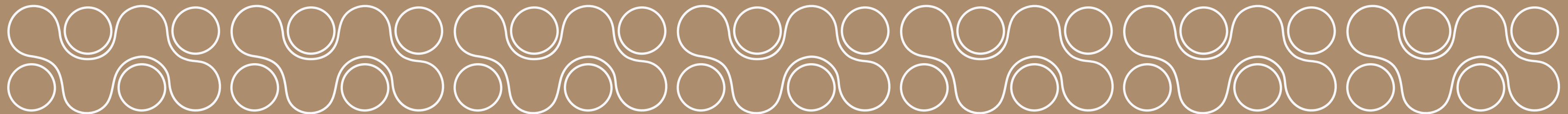
ERD:

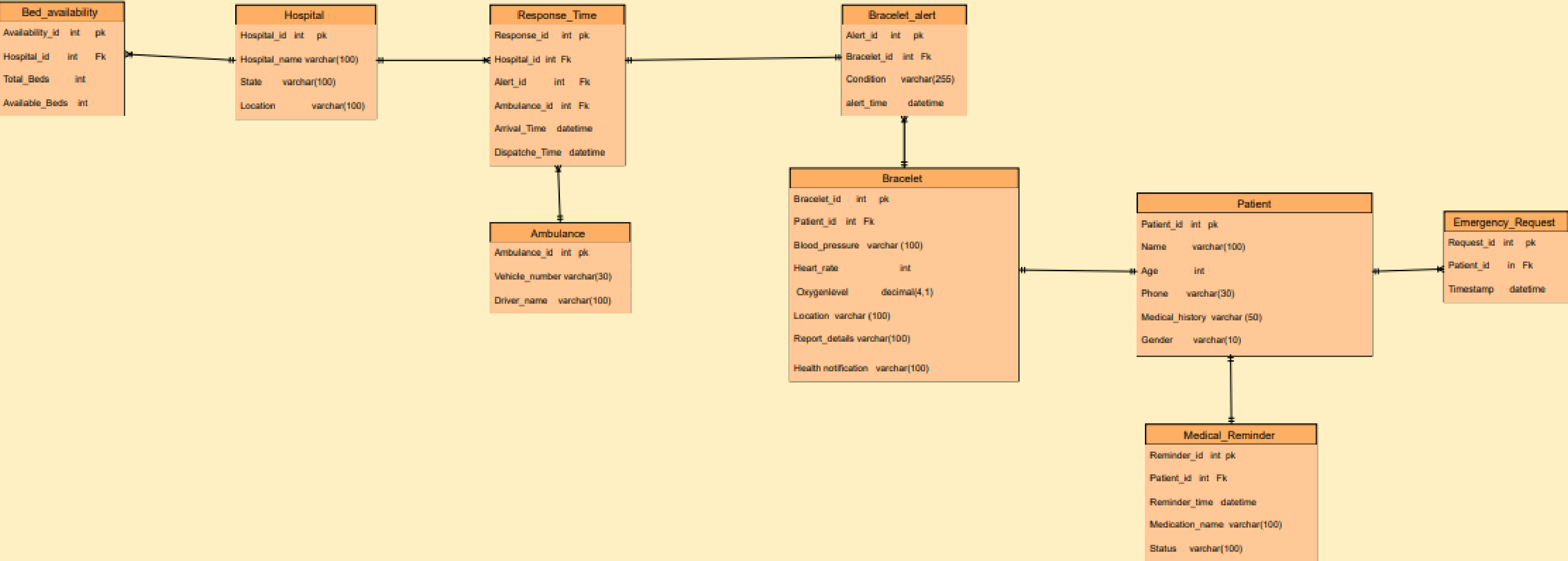
The Entity Relationship Diagram (ERD) illustrates the core structure of the Smart Emergency Medical Response System.

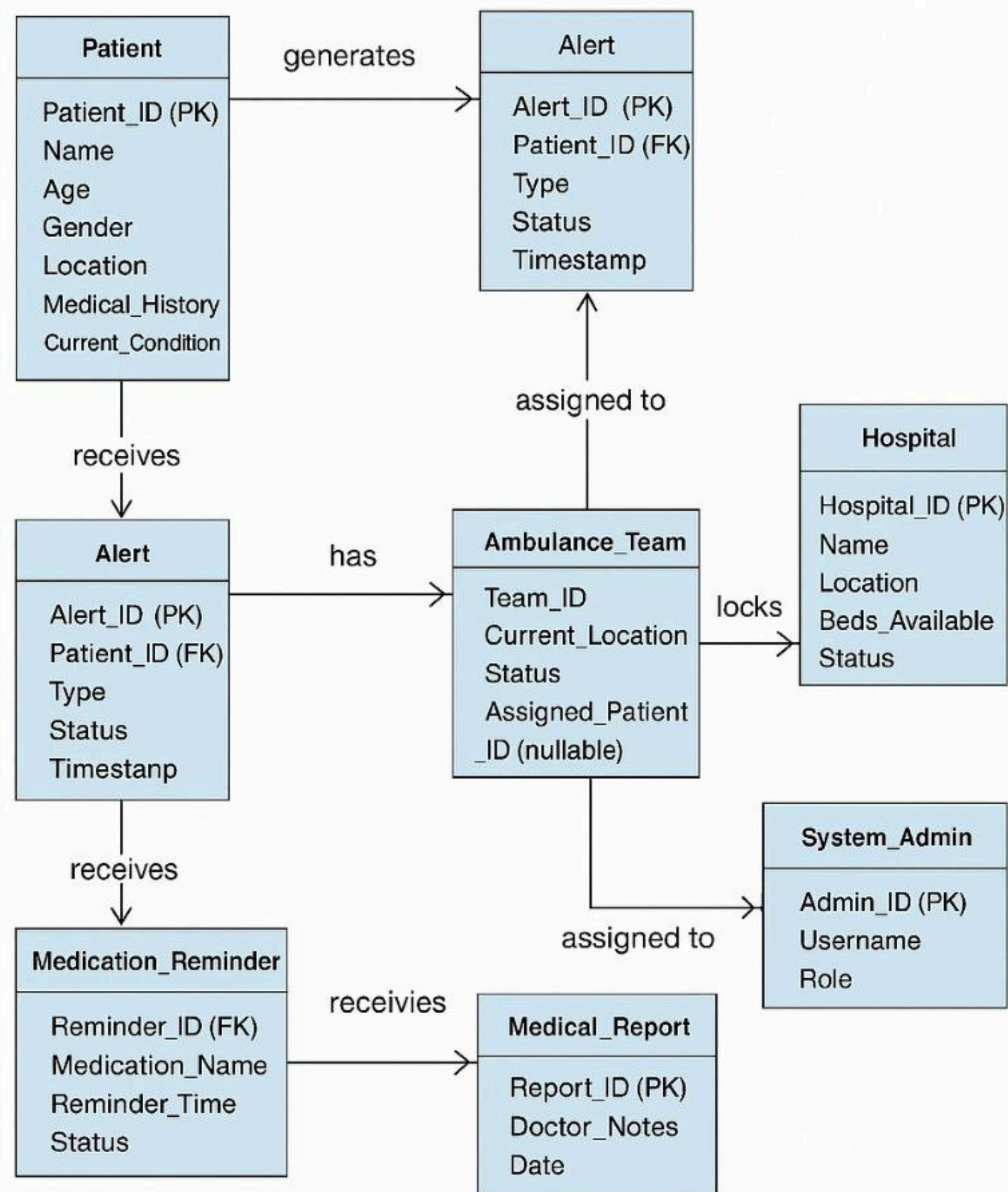
It shows the main entities involved, such as Patients, Hospitals, Vital Signs, and Ambulance Teams, and how these entities interact with each other.

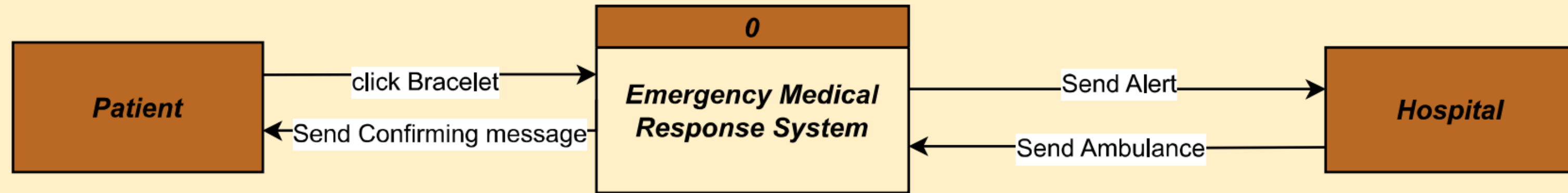
The ERD helps in designing the system's database by organizing data efficiently and showing relationships clearly.

This ensures that patient monitoring, emergency alerts, medication reminders, and hospital coordination are managed accurately and in real time.

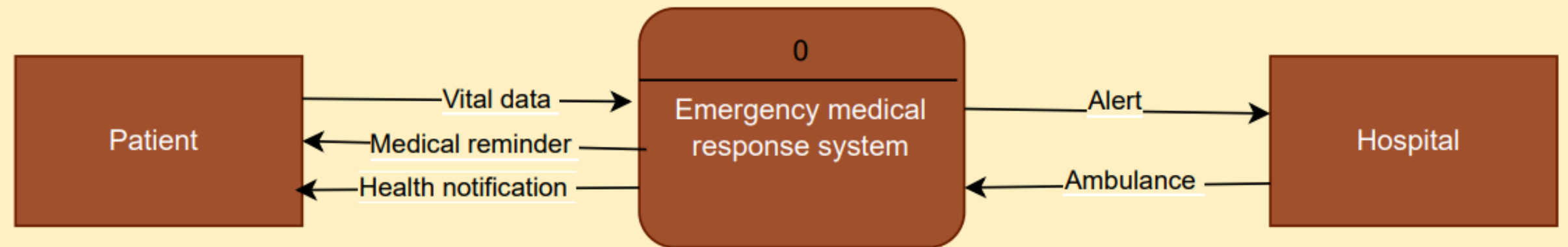


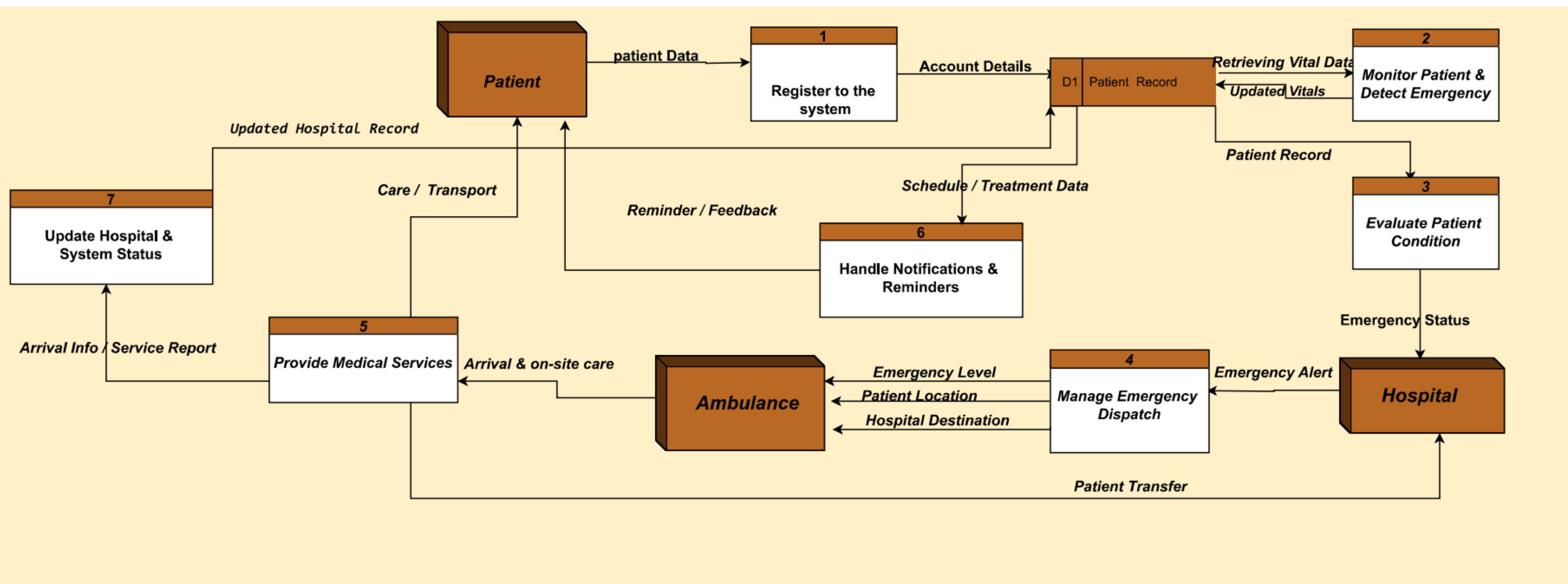


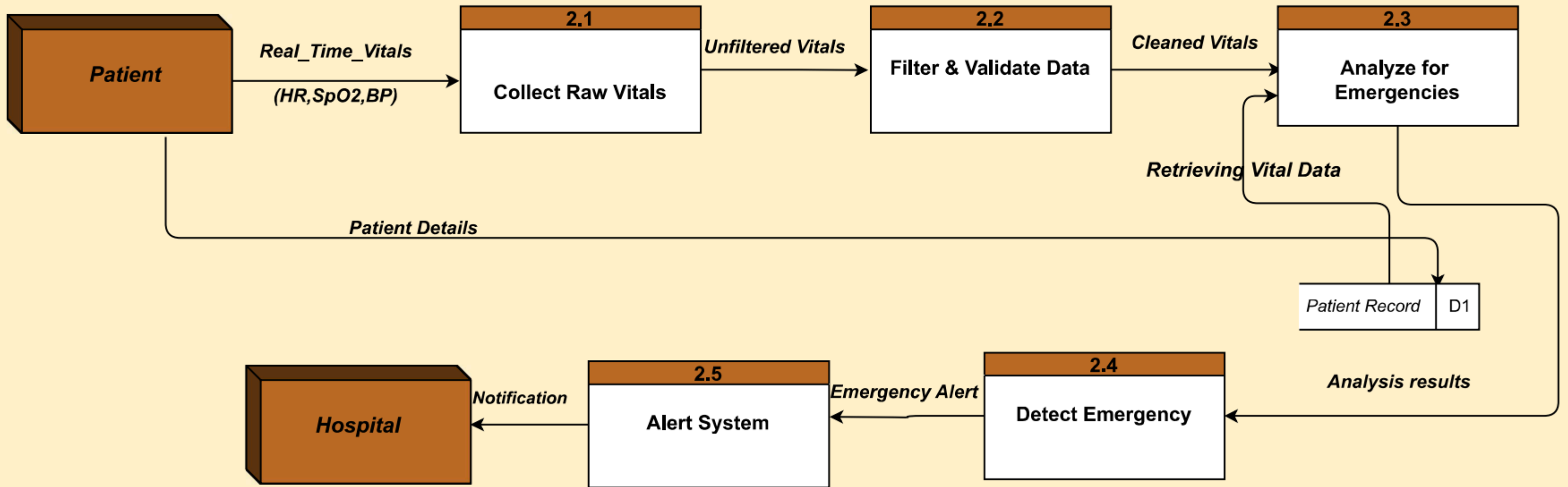




Context :Context Diagram This diagram shows how the Smart Emergency Medical Response System connects with key external entities such as patients, hospitals, and ambulance teams. It outlines the main data flows and interactions that support emergency alerts, patient monitoring, and hospital coordination







thank you

